

Mahmoud Abdel Latif

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SUMMARY

DevOps Engineer focused on cloud infrastructure, building hands-on, end-to-end projects, including a hybrid environment connecting an on-premise server to GCP via Terraform and VPN networking, plus multiple AWS and Kubernetes deployments covering CI/CD, containerization, and observability. Comfortable across Linux administration, Infrastructure as Code, and hands-on monitoring setups, with a consistent habit of documenting and publishing every project end-to-end.

PROJECTS

Hybrid Cloud Infrastructure on GCP | *Terraform, GCP, Tailscale VPN, FastAPI, Nginx* [GitHub](#)

- Designed and provisioned a modular hybrid-cloud architecture on GCP with Terraform, including a custom VPC, Cloud NAT, and a hardened bastion host.
- Connected an on-premise home server to a private GCP subnet via Tailscale VPN, enabling secure hybrid connectivity with no public exposure.
- Deployed a FastAPI backend behind an Nginx reverse proxy, with remote Terraform state stored in GCS to support safe, collaborative infrastructure changes.
- Configured GCP Cloud Monitoring with custom alerting policies for proactive infrastructure observability.

CI/CD Pipeline for Node.js Application on AWS EKS | *GitHub Actions, Docker, ECR, EKS, Kubernetes* [GitHub](#)

- Built an end-to-end CI/CD pipeline that builds, tests, containerizes, and deploys a Node.js application to Amazon EKS using GitHub Actions.
- Automated Docker image build and push to Amazon ECR, with deployment updates rolled out to EKS via a manually-triggered CD workflow.
- Exposed the application through a Kubernetes LoadBalancer Service and enabled CloudWatch logging for runtime visibility.

WordPress Platform on AWS EKS | *EKS, EBS, EFS, Docker, Kubernetes* [GitHub](#)

- Deployed a production-style WordPress and MySQL environment on Amazon EKS with persistent storage architecture.
- Configured EBS volumes for database persistence and EFS shared storage for WordPress uploads across multiple application replicas.
- Created modular Kubernetes manifests for Deployments, Services, Persistent Volumes, Persistent Volume Claims, and Load Balancer exposure.

Stateful Python Application on Multi-Node Kubernetes Cluster | *Docker, Kubernetes, AWS EFS, NGINX Ingress* [GitHub](#)

- Built and containerized a Python web application (user registration, login, and to-do management) and deployed it as a 3-replica Kubernetes Deployment for high availability.
- Persisted application data via Kubernetes PV/PVC backed by AWS EFS through the EFS CSI Driver, giving all replicas ReadWriteMany access to the same shared storage.
- Exposed the application through an NGINX Ingress Controller with HTTP/HTTPS routing, and validated persistence by writing data from one pod and confirming it was readable from another after rescheduling.

EDUCATION

Mansoura University

BEng. in Computer and Communications Engineering

October 2022 – Expected June 2027

Mansoura, Egypt

TECHNICAL SKILLS

Cloud & Infrastructure: AWS (EC2, EKS, ECR, IAM, VPC, Subnets, Security Groups, CloudWatch, EBS, EFS, Load Balancing), GCP (VPC, Cloud NAT, Cloud Monitoring), Terraform.

Containers & Orchestration: Docker, Docker Compose, Kubernetes (kubectl, Deployments, Services, Ingress, PV/PVC, EFS/EBS CSI).

CI/CD & Automation: Jenkins, GitHub Actions.

Monitoring & Observability: Prometheus, Grafana, AWS CloudWatch, GCP Cloud Monitoring.

Networking & Security: TCP/IP, DNS, HTTP/HTTPS, Routing, Load Balancing, VPC Networking, Security Groups, Tailscale VPN.

Operating Systems & Scripting: Linux (systemd, SSH, permissions, package management), Bash.

Version Control: Git, GitHub.